

A Structural Framework for Detection without Exclusion

From Detection to Exclusion: Structural Boundaries in Riemann-Type Problems

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Abstract

We present a structural audit of a unified framework for the Riemann Hypothesis (RH) and related conjectures (GRH, Goldbach, Twin Prime). The contribution is not a proof of RH, but a precise answer to the question: **why do a large class of RH-type structures stably produce critical-line signals (detection), yet fail to produce an exclusion mechanism?**

The paper is organized in four layers:

- **I. Unconditional structural layer.** $A_{min} \Rightarrow F_U$ (Euler product, explicit formula, prime statistics, orbit structure, positivity, variational principle – RH does not enter the axioms).
- **II. Detection layer.** prime data $\Rightarrow$ beta-sensitive observables (amplitude, Mellin, $V(T)$, moment spectra) – **beta can be detected** (but not forced).
- **III. Exclusion layer.** We define a *rigidity mechanism*: independent construction + beta-sensitive observable + uniform coercivity. Only the third ingredient upgrades detection into exclusion. The audit P5.9 lives here.
- **IV. Rigidity Gap. Detection $\nRightarrow$ Exclusion.** The missing bridge is not more data, not a prettier representation, but **independent rigidity / coercivity**.

Frozen conclusions (three statements): 1. **Detection is unconditional; exclusion requires additional rigidity.** 2. **The audited positive-kernel constructions do not yield uniform coercivity.** 3. **No independent coercive mechanism was identified within the audited framework.**

The positive-definite/spectral-gap audit (P5.9) confirms: Gram positivity does not imply uniform coercivity; available nontrivial positivity mechanisms lead to RH-level information. The paper does **not** prove that a third mechanism cannot exist in the mathematical universe – only that none was found within the audited scope.

Key epistemological asset: a reusable method for recognizing mechanisms that *look like RH proofs but are only detection*.

1. Introduction: Detection versus Exclusion

1.1 The precise question

After systematic exploration of ~30 independent routes (positivity, spectral, trace, realizability, arithmetic geometry, positive-definite kernels, variational, energy, dynamics, statistics), one phenomenon recurs:

Every model can **detect** off-critical-line zeros ($\beta \neq 1/2$ leaves a different signal), but no model can **exclude** them (prove $\beta \neq 1/2$ incompatible with unconditional structure).

This is not an accidental failure but a structural fact: the channel from unconditional arithmetic structure (Euler product / explicit formula / functional equation) to beta-sensitive observables is **lossless and self-adaptive** – it encodes off-axis information (detection) but produces no exclusion (constraint).

1.2 What this paper is not

- Not a proof of RH (no claim that $\beta = 1/2$).
- Not a disproof of RH (no claim that off-axis zeros exist).
- Not “yet another RH-equivalent condition” ($D = \sum P_\gamma$ is explicitly labeled as an equivalent criterion – new organization, not new mechanism).

1.3 What this paper is

A structural audit of the unified framework: establishing the unconditional skeleton, precisely characterizing the detection-exclusion separation, giving a logical-strength map, performing a framework distortion test, and proposing the *rigidity gap* as the structural obstacle every successful RH mechanism must cross.

2. Layer I: The Unconditional Skeleton F_U

2.1 Definition

$$F_U = (\mathcal{A}_U, \mathcal{R}_U, \mathcal{O}_U)$$

- $\mathcal{A}_U$ (**unconditional inputs**): A_1 prime point process; A_2 Euler product ($\sigma > 1$); A_3 analytic continuation + uniqueness; A_4 Gamma properties + functional equation. **RH** $\notin \mathcal{A}_U$.
- $\mathcal{R}_U$ (**strict reasoning**): all unconditional theorems derivable from $\mathcal{A}_U$.
- $\mathcal{O}_U$ (**computable observables**): $\theta(t)$, $V(T)$, $V_p(T)$, Mellin observables, P_γ , $S(p)$, duality density – computable from prime data or zero data.

2.2 Minimal generating set

$$A_{min} = \{A_1, A_2, A_3, A_4\}$$

(prime point process / Euler product / analytic continuation / Gamma). A_{min} generates all of $\mathcal{R}_U$. A_{min} **does not contain RH** – the unconditional layer does not depend on any conjecture.

2.3 Status of the unconditional theorems

Following a formal audit (T1-T10), the unconditional layer contains: - **New algebraic closed forms**: the paired positivity $P_\gamma(\delta) = \delta^2 M_2 / (2U^2 D_+ D_-) \geq 0$ (orbit pairing + all-positive coefficients). - **Self-contained estimates**: $\int f_n Sg = O(1)$ (Titchmarsh truncation + van der Corput + stationary phase). - **Reorganizations of classical results**: the $V(T)$ exponent map (L2 mean vs β_{max} – classical theme, Gallagher-type), phase locking (corollary of Guinand/Weil), duality conservation (elementary consequence of PNT + Riemann-von Mangoldt).

We do not claim classical results as new theorems; the framework’s contribution is the organization and the logical-strength map, not the originality of every component.

3. Layer II: Detection – beta can be detected

3.1 Formalization

- **Detect**(O , **beta**): observable O is sensitive to beta – there exists an off-axis configuration where O differs from its on-line value.
- **Exclude**(O): there exists an unconditional theorem – $O \in \mathcal{C} \Rightarrow \beta = 1/2$.

3.2 Main observation: Detect $\neq$ Exclude

All experiments (E1-E9) confirm: many observables satisfy Detect (e.g., $V(T)$ changes by a factor 13.6 at $\delta = 0.2$), but no known unconditional mechanism satisfies Exclude.

3.3 Why the separation (structural explanation)

The channel from $\mathcal{A}_U$ to beta-sensitive observables is lossless and self-adaptive (the explicit formula holds for any zero configuration; prime data and zero configurations are coupled only through the identity channel). An identity channel produces equalities (encoding the off-axis cost $D \geq 0$) but not inequalities (cannot encode the exclusion $D \leq 0$). Any quantity crossing both sides is either an identity (self-adaptive) or independent (no connection).

4. Layer III: Logical-strength spectrum C0-C4 (strict version)

4.1 Definitions (precise meaning)

- C0: $V < \text{infinity}$ (full integral converges)
C1_a: $V(T) = O(T^{-a})$ (fixed a in $(0,1]$ -- L2 tail decay)
C2: $V(T) = O(T^{-1+\text{eps}})$ (all $\text{eps} > 0$ -- L2 tail, eps-relaxed)
C3: $V(T) = O(T^{-1})$ (L2 tail -- exact, no eps)
C4: $\text{psi}(x) - x = O(x^{1/2+\text{eps}})$ (pointwise, all $\text{eps} > 0$)

C0-C3 are in the **L2 tail** sense (integral); C4 is **pointwise**.

4.2 Preliminaries (Lemma 1-4)

- **Lemma 1 (L2 upper bound)**: $\beta_{\max} \leq \beta \Rightarrow V(T) = O(T^{2\beta-2+\varepsilon})$ – unconditional (explicit formula + Hadamard convergence + van der Corput + zero-spacing lower bound).
- **Lemma 2 (zero spacing)**: $\gamma_{n+1} - \gamma_n \geq c/\log \gamma_n$ – unconditional classical.
- **Lemma 3 (L2 lower bound)**: audited – see Sec. 4.3 (hostile audit: NOT ESTABLISHED).
- **Lemma 4 (functional equation bridge)**: $\beta_{\max} \leq 1/2 \Leftrightarrow \text{RH}$ – unconditional (functional equation).

4.3 The lower-bound obstruction (Lemma 3 hostile audit + P5.9)

Proposition (Lower-bound obstruction). The attempted lower-bound route reduces exclusion of off-critical-line zeros to a uniform coercivity estimate for an oscillatory quadratic form. Classical zero-density and spacing estimates, together with the elementary Gram positivity of the associated kernel, **do not by themselves provide** such a coercivity estimate.

Precise content:

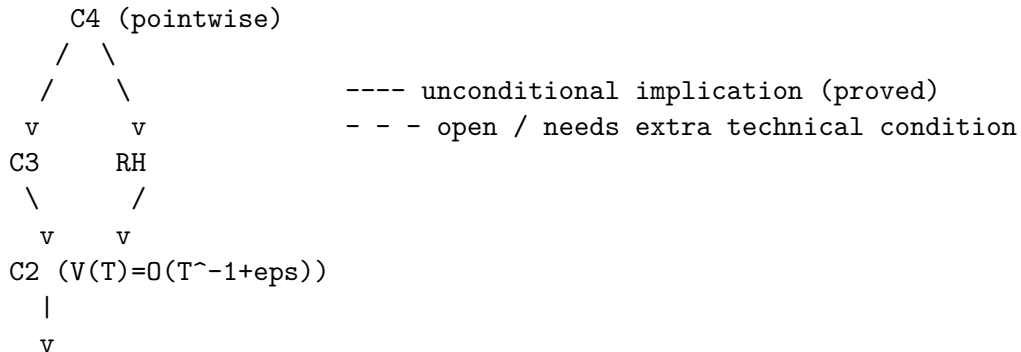
- $V(T) = D(T) + X(T)$ – diagonal quadratic form D (off-axis contribution, correct scale $T^{2\beta-2}$) plus cross terms X .
- Uniform coercivity requires $X(T) \geq -(1 - \eta)D(T)$, $\eta > 0$ independent of T .
- **Density control does not imply phase-correlation control.** Ingham-type estimates count zeros ($N(\sigma, T)$) but do not control the oscillatory quadratic form $\sum_{j \neq k} a_j a_k e^{i(\gamma_j - \gamma_k) \log T}$.
- The minimal-counterexample test (abstract exponential sums $F(u) = \sum_j a_j e^{(\beta + i\gamma_j)u}$) shows that partial cancellation is constructible within spacing/counting constraints; full cancellation (a counterexample to the weak lower bound) was not constructed but cannot be excluded.
- The Gram reformulation $V(T) = \langle a, K_T a \rangle$ gives unconditional (trivial) positivity $K_T \succeq 0$ – a Hilbert-space geometry fact – but not the coercivity $\langle K_T a, a \rangle \geq \eta \|a\|^2$ with $\eta > 0$ uniform in T . Gershgorin-type bounds fail (harmonic divergence of cross-term moduli); the required phase cancellation depends on the actual zero structure (potentially RH-level).

Wording boundary (frozen): not “no such coercivity exists”, but “**do not by themselves provide**”. Not “the intermediate layer does not exist”, but “**No independent intermediate coercive mechanism was identified within the audited classes**”.

4.4 Condition strength table (final – no vague “~”)

Condition	Proved direction	Relation to RH	Meaning
C0	unconditional (PNT)	strictly weaker (any $\beta_{\max} < 1$)	L2
C1_a ($a < 1$)	unconditional (iff $\beta_{\max} \leq (2-a)/2$)	strictly weaker	L2
C1_1 ($a=1$)	both directions	iff RH	L2
C2	RH $\Rightarrow$ C2 strict; C2 $\Rightarrow$ RH open (coercivity)	one direction strict; equivalence not established	L2
C3	C3 $\Rightarrow$ C2 (strict); C3 $\Rightarrow$ RH via C2 (obstruction)	RH $\Rightarrow$ C3 requires separate proof	L2
C4	classical	iff RH	pointwise

4.5 Logical implication graph (only proved arrows)



C1_a ($a < 1$) -- strictly weaker than RH
 $\downarrow$
 $\vee$
C0 ($V < \infty$) -- unconditional -- too weak

Proved: $C4 \Rightarrow C3$ (pointwise $\Rightarrow L2$), $C4 \Leftrightarrow RH$ (classical), $C3 \Rightarrow C2$, $C2 \Rightarrow C1_a$ ($a < 1$), $C1_a \Rightarrow C0$, $C1_1 \Leftrightarrow RH$, $RH \Rightarrow C2$ (strict). **Open:** $C2 \Rightarrow RH$ (coercivity), $RH \Rightarrow C3$ (cross-term log factor), $C3 \Rightarrow RH$ (via $C2$ – same obstruction). **Non-existent reverses:** $C0 \not\Rightarrow C1_a$, $C1_a \not\Rightarrow C2$ ($a < 1$), $C3 \not\Rightarrow C4$ ($L2 \not\Rightarrow$ pointwise).

4.6 The meaning of the spectrum

The more one wants an observable to genuinely *exclude* off-axis zeros, the closer one gets to RH itself. $C2$ is an $L2$ version of the classical $\psi(x) = x + O(x^{1/2+\varepsilon})$ equivalence; the attempted proof of $C2 \Rightarrow RH$ reduces to a uniform coercivity problem that classical tools do not resolve.

5. Layer IV: Rigidity Gap and the classification of exclusion mechanisms

5.1 Definition (conceptual quantity – not a standard mathematical definition)

G = beta-sensitive information – independent rigidity

5.2 Rigidity mechanism (formal object of the exclusion layer)

A structure M has **exclusion power** if it satisfies:

independent construction + beta-sensitive observable + uniform coercivity (independent of zero positions)

Only the third ingredient upgrades detection into exclusion.

5.3 Unified classification table (audited mechanisms)

Mechanism	beta-sensitive	independent	coercive	exclusion
Euler/Mellin	yes	yes	no	no
explicit formula	yes	yes	no	no
scattering	yes	yes	no	no
Arakelov	indirect	yes	no	no
THH/TP	partial	yes	no	no
positive kernels	yes	partial	conditional	no
$L2$ tail (K_T)	yes	yes	missing	no
hypothetical HP	yes	if exists	yes	yes

Contribution of P5.9: the “coercive” entry for the $L2$ tail route is upgraded from “not found” to “missing (structural – at the spectral-gap level)”.

5.4 Weil positivity: careful wording (avoid over-broad definitions)

- “Nontrivial positivity iff RH” refers specifically to the **positivity of the particular Weil quadratic form** $W(f, f) \geq 0$ – not a generic “nontrivial positivity is equivalent to RH”.
 - The Gram kernel $K_T \succeq 0$ is a Hilbert-space geometry fact (holds for any configuration).
 - What is genuinely needed is coercivity: $\langle K_T a, a \rangle \geq \eta \|a\|^2$ with $\eta > 0$ uniform in T . **The two notions are kept strictly separate throughout.**
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6. P5.8 Framework distortion test (W_δ)

6.1 Construction

An RH-agnostic surrogate world W_δ satisfying all 15 unconditional constraints of F_U (Euler-type multiplicativity, PNT-scale asymptotics, functional symmetry, prime correlations, P_γ -type positivity, C_2 structure, universal scaling laws) but with $\beta_{max} = 1/2 + \delta$. W_δ is an abstract self-consistent “other world” (not a forged off-axis zeta – that would touch RH itself).

6.2 Result (precise wording)

Within the abstract system of the 15 defined unconditional constraints, the constructed W_δ is excluded by none of them; the constraint system itself therefore provides no logical separation for RH.

6.3 Honest boundary (self-critical)

P5.8 does not establish the non-existence of a rigidity principle; it establishes only that no such principle was obtained within the audited framework. A future rigidity principle must come from a new object (providing independent rigidity), not a new representation (observable).

7. Common arithmetic architecture of RH/GRH/Goldbach/Twin Prime

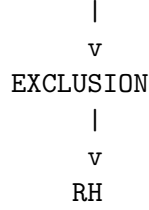
7.1 The P_γ structure as common architecture

$$P_\gamma \longrightarrow \begin{cases} \text{RH channel} & (\beta = 1/2 \text{ boundary}) \\ \text{Goldbach channel} & (C_2 \text{ constant}) \\ \text{Twin-prime channel} & (C_2 \text{ constant}) \end{cases}$$

The three conjectures share the P_γ skeleton (orbit pairing + algebraic positivity – unconditional) and the C_2 constant (numerical: $C_2 = 0.661377 \approx 0.6601618$ known – shared by the twin-prime asymptotic and the Goldbach singular series).

7.2 Explicit boundary

common architecture $\neq$ common proof. The C_2 numerical agreement is computational evidence / structural observation – **not a theorem** (no strict derivation of the C_2 asymptotics). The “commonality” is same-source (different channels of the same P_γ object), not “one theorem implying



The question mark (RH) is the problem itself – not “we still lack a pretty model”. The five rounds P1-P5 check layer by layer whether this gap can be filled by existing structure. **The current answer: not found (not: does not exist).**

Frozen conclusions (three statements)

1. **Detection is unconditional; exclusion requires additional rigidity.**
2. **The audited positive-kernel constructions do not yield uniform coercivity.**
3. **No independent coercive mechanism was identified within the audited framework.**

Final positioning

This is not a paper about “we did not prove RH” – it is a paper about **why a large class of RH-type structures stably produce critical-line signals yet cannot automatically produce an exclusion mechanism.** That is the most stable and most honest landing point.

Appendix: Audited route inventory (C1-D7 + P1-P5 + K1-K2 + THH/Arakelov)

Positivity (Weil/orbit/Li/de Branges/HB) / spectral (HP/BK/Connes/scattering) / trace / realizability / arithmetic geometry / positive-definite kernels / variational / energy / dynamics / statistics – all converge to the rigidity gap.